

Prediction Model for Defects in Lead and Lead-free Aprons

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Abstract—Personal radiation protective equipment (PRPE) is prone to defects in the attenuating layers, resulting in inadequate protection. Hence, quality control (QC) of PRPE is needed to assess its integrity. Unfortunately, QC of PRPE is laborious and time consuming. This study aimed to predict the QC outcome of PRPE without x-ray imaging based on readily available predictors. PRPE QC data of a general hospital from 2018 to 2023 was used for both prediction models based on logistic regression and random forests (RF). The data were divided into a training set containing all data from 2018 to 2022 and a holdout set containing the data from 2023. The predictors were brand, age, size, type, visual defects, and department. The prediction performances were compared using confusion matrices and visualized with receiver operating characteristic (ROC) curves. Prediction accuracies of at least 80% were achieved. Further model tuning especially improved the RF model to a precision up to 97% with a sensitivity of 80% and specificity of 86%. All predictors, except visual defects, significantly impacted the probability of passing. The predictor brand had the largest contribution to the predictive performance. The difference in pass probability between the best-performing and the worst-performing brand was 35.1%. The results highlight the potential of predicting PRPE QC outcome without x rays. The proposed prediction approach is a significant contribution to an effective QC strategy by reducing time consuming x-ray QC tests and focusing on garments with higher probability of being defective. Further research is recommended.

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Key words: occupational safety; quality control; radiation protection; statistics

INTRODUCTION

OPERATORS USING x-ray systems should employ appropriate measures to shield themselves from scattered radiation. Especially during fluoroscopy-guided procedures (FGP), where

proximity to the patient is inevitable, personal radiation protective equipment (PRPE) is crucial to reduce occupational exposure (Theocharopoulos et al. 2006; Klein et al. 2018; Crowhurst et al. 2021). PRPE consists of lead, lead-composite, and lead-free aprons, vests, skirts, and thyroid shields that are expressed in terms of lead equivalence. Lead equivalence states to what thickness of pure lead the lead-composite or lead-free material corresponds concerning x-ray attenuation (Yaffe et al. 1991; Peters et al. 2017; Aral et al. 2020). The most common lead equivalence is 0.25 mm, resulting in 0.50 mm when used in a wrap-around garment, defined and determined by the 61331-3:2014 standard of the International Electrotechnical Commission (IEC) for radiation protection against x rays up to 150 kV during radiological and interventional procedures (Eder and Schlattl 2018).

Unfortunately, PRPE is prone to deterioration in the attenuating layers (Glaze et al. 1984; Oppliger-Schäfer and Roser 2009; Bawazeer 2021; Kellens et al. 2022). The defects, mostly in the form of holes and tears along the seaming, are invisible to the human eye as the attenuating layer is typically covered by the outer fabrics. X-ray imaging is thus required to visualize and assess the defects in PRPE (Kellens et al. 2022). Generally, quality control (QC) of PRPE is performed on an x-ray table using fluoroscopy. This allows the inspector to scan through the garment. Nonetheless, QC of PRPE is laborious and time consuming, as each piece must be scanned thoroughly while qualifying and quantifying each encountered defect. In addition, fluoroscopically scanning each piece leads to high fluoroscopy times, possibly resulting in increased radiation exposure for the QC surveyor.

Therefore, this study aims to develop a prediction model that is capable of predicting the QC outcome based on qualitative features available without x-ray imaging. This could potentially lead to a more effective QC policy subjecting the garments that are at higher risks to more frequent QC. More specifically, this study aims to predict the QC outcome of a garment with a lead equivalence of 0.50/0.25 mm and to identify the most influential predictor. Thyroid shields are deliberately left out as they are less prone to defects due to their size and could therefore bias the model (Kellens et al. 2022).

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MATERIALS AND METHODS

Dataset collection and pre-processing

All PRPE of a general hospital were inspected yearly from 2018 to 2023. Each piece of PRPE was tagged with a unique ID number (TN) in order to have a chronological record of every tear and test result through the years. The QC was performed by medical physics experts (authors P.K. and A.D.H.) on behalf of the health and safety department of the hospital. Each QC of a piece of PRPE consisted of a preliminary visual inspection. The outer fabric of the garment as well as the closure of the buckles, hook-and-loop fasteners, and magnets were examined. Any damage to the exterior surface was documented for each piece of PRPE. Following the visual inspection, an x-ray-based integrity analysis was performed with a Siemens Luminos dRF (Siemens Healthineers, Erlangen, Germany) using fluoroscopy in a standardized anterior-posterior abdomen protocol with a field size of 42 cm × 42 cm and a source-to-image distance of 115 cm. Each encountered defect was recorded and estimated using an x-ray opaque grid of 1 cm accuracy. The applied center’s rejection criteria were based on the criteria of Lambert and McKeon (2001). In brief, total defect areas larger than 670 mm² lead to a rejection. In addition, an institutional rule was applied that a single tear of more than 15 mm² would also result in rejection regardless of the location of the tear in the garment. For each piece, the following data, if available, were collected: (1) department of use, (2) size, (3) type, (4) brand, (5) manufacturing date, (6) lead equivalency, (7) visual defects, and (8) QC result. The value of the outcome, QC result, is either pass or reject corresponding to the value of 1 or 0, respectively.

Observations with a lead equivalence other than 0.50/0.25 mm or observations regarding thyroid shields were likewise deleted. The dataset was divided into a training set containing all data from 2018 to 2022 and a holdout set containing the data from 2023. The qualitative predictors that were included in the predictive models are shown in Table 1, together with their possible data values.

Data analysis, modeling, and performance

Univariate descriptives are given as mean ± standard deviation (SD) for continuous variables or as percentages with the corresponding absolute value in parentheses for categorical variables. Missing data proportions are reported as percentages. Associations between categorical variables and the outcome variable were tested two-sided using the chi-square test of independence with degrees of freedom equal to the number of categories −1. Reported p-values are two-sided on a 5% significance level. Uncertainty around the regression coefficients is reported as 95% confidence intervals.

Two approaches were used to predict the QC result, being (1) logistic regression and (2) the machine learning algorithm random forests (RF). In the logistic model, categorical variables with *k* > 2 categories were coded into *k* − 1 extra dummy variables with possible values of 0 or 1 with the first category as reference. This facilitates the interpretation of the main effect of different levels of the categorical predictor on the odds of passing a QC. The log odds of passing the QC for the reference category is simply the intercept in the logistic regression model. The regression coefficients of the *k* − 1 dummies reflect the ratio of the odds of passing a QC for each level of the categorical predictor compared to the reference level on the logarithmic scale. While the logistic regression model assumes linear effects of continuous predictors on the logarithmic scale, no linearity is imposed for RF. RF creates a different training subset from sample training data with replacement, which results in multiple classification trees for the probability of passing. The RF algorithm then obtains a majority vote of the trees as an average (Breiman 2001).

Validation of the logistic regression and random forest is performed by comparing the predictions of the models on the holdout set with the actual QC results. The model performances will be compared and visualized using receiver operating characteristic (ROC) curves. These curves display the sensitivity against the specificity and allow assessment of the tradeoff between both rates. In addition, ROC curves allow visualization of threshold tuning (the default

Table 1. Predictor description and possible values. Binary variables were dummy coded with the dummy coded value in parentheses.

Predictor	Description	Categories
Age	The age of the piece in years counting from the manufacturing date.	Discrete numeric variable
Size	An ordinal variable representing the size of the garment.	Extra small < small < medium < large < extra large
Visual defects	A binary variable whether the garment has a visual defect.	Yes (1) or No (0)
Type	A nominal variable representing the PRPE model types.	Apron, vest, and skirt
Department	A nominal variable representing the department of use of the PRPE.	pain clinic (PC), cardiology, surgery theatre (ST), medical imaging (MI), gastroenterology, oral and maxillofacial surgery, and urology (G/MKA/URO)
Brand	A nominal variable representing the brands of PRPE.	Brand A (Infinity), Brand B (Tema), Brand C (Scanflex) Brand D (Protec X), Brand E (MDT) and Other (all less frequent brands pooled together)

Table 2. Associated measures of a confusion matrix.

Measure	Formula
Sensitivity	$\frac{TP}{TP+FN}$
Specificity	$\frac{TN}{TN+FP}$
False positive rate (FPR)	$\frac{FP}{FP+TN}$
Precision	$\frac{TP}{TP+FP}$

threshold for interpreting probabilities to class labels is 0.5, where all values equal to or greater than the threshold are mapped to pass and all other values to fail for examples) toward fewer false positives, i.e., a falsely predicted pass. An associated statistic to ROC curves is the area under the curve (AUC) indicating how efficient the model is in distinguishing between passing the QC and failing the QC. For each model, a confusion matrix, i.e., a tabular summary of the classification (in)accuracies, is displayed as counts of true positives and negatives. The definition of the associated measures being (1) sensitivity, (2) specificity, (3) false positive rate (FPR), and (4) precision, as shown in Table 2.

Software

All statistical analysis was performed in R (team Rc, Vienna, Austria) using Rstudio (Rstudio Team 2020, PBC, Boston, MA). RF modeling was performed using the random forest package (Liaw and Wiener 2002). Missing values were multiply imputed by chained equations (MICE) (van Buuren and Groothuis-Oudshoorn 2011).

RESULTS

Exploratory data analysis and univariate descriptives

Overall, 1,489 unique pieces of PRPE (individual TN) were inspected, resulting in a total of 4,482 QCs. The original dataset was reduced to 2,504 QCs (741 unique TN) after the exclusion of thyroid shields and pieces with lead equivalences other than 0.50/0.25 mm. In total, 22.5% (564) QCs led to a rejection.

Age had a mean value of 3 (± 2) y with a maximum value of 13 y. The association for the categorical predictors with the outcome variable is summarized in Table 3. The data values in this table are for each unique TN. Note that the data values for size do not add up to 741 as size had missing values. Brand and size yield significant marginal associations with QC result.

The observed probabilities of pass for each brand category are shown in Fig. 1. Pieces of brand A were rejected in 36.9% (416 pieces) of all QC performed over the study period while brand E only rejected 1.8% (12 pieces) during the study period. In a next step we investigate the simultaneous effect of all predictors on QC using logistic regression.

Predictive modeling of QC result

The logistic regression equation of the full model was:

$$\text{logit}[\pi(x)] = \beta_0 + \beta_1 \text{Department} + \beta_2 \text{Brand} + \beta_3 \text{Type} + \beta_4 \text{Visuals} + \beta_5 \text{Age} + \beta_6 \text{Size}, \quad (1)$$

with $\pi(x)$ being the probability of passing the QC. The reference category for each treatment coded predictor used in this model are shown in italics in Table 3. The most important predictor to influence the odds of passing the QC was brand (p-value < 0.001). Especially the PRPE of brand E significantly (p-value < 0.001) increases the odds of passing the QC as compared to brand A. The odds of a skirt from brand E to pass the QC is 171 times the odds of a skirt from brand A to pass the QC.

Table 4 displays the confusion matrix of the fitted logistic regression model tuned toward a threshold corresponding with less falsely predicted QC passes. The accuracy was 80%.

Table 5 displays the confusion matrix of the random forest approach tuned toward a threshold corresponding with fewer falsely predicted QC passes. The accuracy was 81%. Fig. 2 displays the ROC curves for both models.

Table 3. Association between the categorical predictors and the outcome variable QC result.^a

Predictor		QC result		ϕ^2
		Rejected	Pass	
Size	<i>XS</i>	3	0	69.70 (***)
	S	18	8	
	M	38	21	
	L	56	192	
	XL	13	4	
Visual defects	<i>No</i>	223	341	1.91 (ns)
	Yes	81	96	
Type	<i>Skirt</i>	124	170	0.94 (ns)
	Vest	126	177	
	Apron	54	90	
Department	<i>Cardiology</i>	75	144	7.3 (ns)
	G/MKA/URO	10	10	
	MBV	35	53	
	ST	155	199	
	PC	29	31	
Brand	<i>Infinity</i>	203	65	348 (***)
	MDT	12	304	
	Protec X	11	18	
	Scanflex	12	20	
	Tema	51	18	
	Other	15	12	

^a PC = Pain clinic, Cardiology = Catheterization lab, ST = Surgery theatre, MI = Medical Imaging, G/MKA/URO = Gastroenterology/Oral and maxillofacial surgery/Urology. The significance codes are: ns > 0.05 > * > 0.01 > ** > 0.001 > ***. Significant results are in bold.

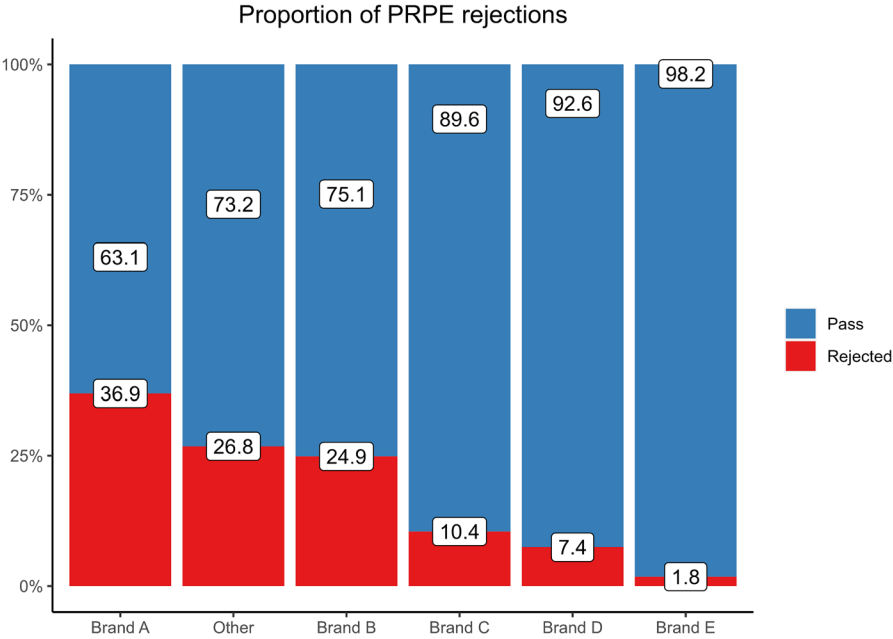


Fig. 1. Proportional view of QC outcome for the five most common brands in increasing order from left to right.

Considering the ROC curves in Fig. 2, both approaches perform quite similarly for different probabilities of classifying a piece of PRPE to pass the QC. However, the advantage of the RF model over the logistic regression model is clear when the associated measures of the confusion matrices (Table 4 and Table 5) are compared. The RF model has a lower FPR, a higher precision, and a higher accuracy without sacrificing sensitivity.

Likewise with the logistic regression, the variable brand is the most important predictor. Fig. 3 displays one of the decision trees of the RF where the influential role of the variable brand is apparent; as for brands E, D, and C, the algorithm already votes a QC pass regardless of the other variables.

Between both approaches, there were 49 contradicting predictions in the holdout set for the same decision boundary. The lower FPR of the RF model is clear when the false positive predictions are checked for these contradicting predictions. Of these 49 contradictions, the logistic regression model made 14 false positive predictions, while the RF model made 10 false positive predictions. Table 6 elaborates these contradicting false positive predictions.

DISCUSSION

PRPE is crucial in decreasing the ionizing radiation exposure of operators using x-rays (McCaffrey et al. 2007; Duran et al. 2011; Klein et al. 2018 Crowhurst et al. 2021). Subsequently, the attenuating layer must be of pristine quality to ensure adequate x-ray shielding. However, recent studies confirm that this is certainly not always the case. X-ray based integrity checks are hence needed to safeguard

the shielding properties of PRPE (Finnerty and Brennan 2005; Bawazeer 2021; Kellens et al. 2022).³ Unfortunately, x-ray-based quality checks are laborious and time-consuming, causing them to be organized only yearly, leaving room for defects to remain undetected for a substantial period.

Therefore, this study aimed to predict the quality control outcome of PRPE using readily available predictor variables to alleviate the laborious quality assurance program of PRPE in large hospitals. Additionally, this study investigated the most important predictor to support further research in personal radiation shielding. Also, the benefit of using a nonlinear nonparametric classification RF model over a linear parametric model is demonstrated. To the best of our knowledge, this kind of work is unprecedented.

For this purpose, a dataset comprising of 5 y of PRPE quality control data from a general hospital was used. Our study indicates that it is feasible to predict the QC outcome of PRPE based on readily available predictors without the

Table 4. Confusion matrix of the logistic regression model.

	Actual rejection	Actual pass
Predicted rejection	62	80
Predicted pass	16	342
Sensitivity	81%	
Specificity	79%	
FPR	21%	
Precision	95%	

³Oppliger-Schäfer D, Roser HW. Quality assurance of x-ray protection clothing at the University Hospital Basel. Annual Conference of SSRMP. 2009.

Table 5. Confusion matrix of the RF model.

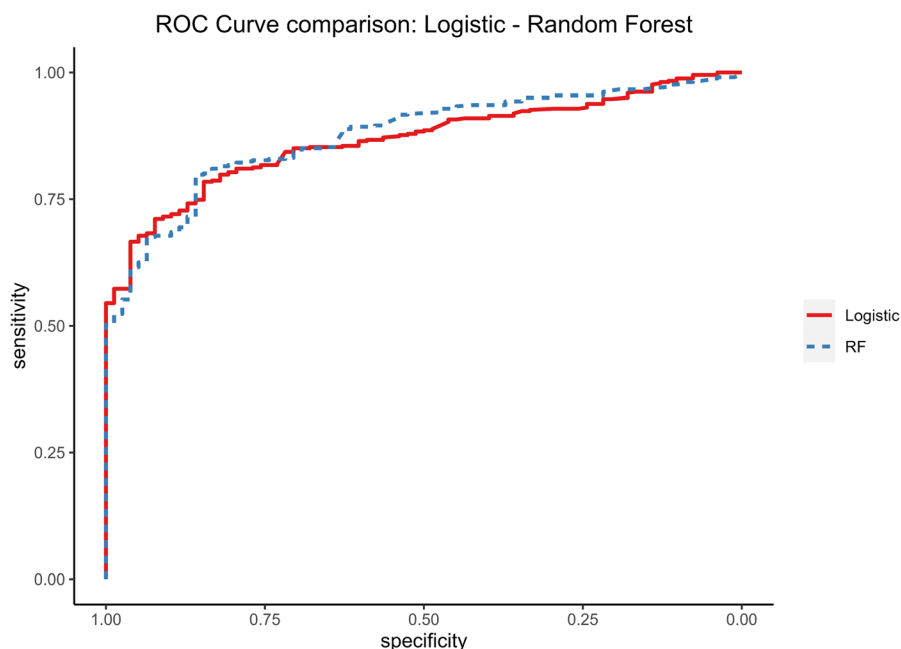
	Actual rejection	Actual pass
Predicted rejection	67	85
Predicted pass	11	337
Sensitivity	80%	
Specificity	86%	
FPR	14%	
Precision	97%	

need of x-ray imaging. Both models achieved an accuracy of at least 80% and a precision of at least 95%. The models are both much better than a random guess and exhibit an AUC of 87% (Fig. 2). It is interesting to know that the AUC is equivalent to the Mann-Whitney U statistic for the median difference between prediction scores for the binary QC outcome (Hanley and McNeil 1982).

Tables 4 and 5 display the advantage of training a non-linear nonparametric RF model over the linear parametric logistic regression in this study setup. The RF model has a lower FPR and a higher precision, inevitably leading to an increased specificity; overall, a better classification performance. This is reflected in the lower number of false positive predictions (10) in the contradicting predictions compared with the false positive predictions (14) from the logistic regression model (Table 6). After all, it is essential for this prediction purpose to keep the false positives as low as possible as these predictions might lead to passing a piece of PRPE that in reality would be rejected. The RF model was chosen as it is nowadays easily available in open-source software and because of its ability to easily handle categorical predictors. Contrasting both models, the parametric logistic regression model assumes

that the probability of a QC pass changes linearly on the logit scale. If a QC program would incorporate a model to facilitate the PRPE QC, the authors suggest a nonparametric nonlinear model to allow more flexibility in the classification based on readily available predictors. Moreover, the model can be tuned toward a minimum of false positives with the trade-off that there will be more false negative results. Despite the better results of the random forest model, false positives were still made. Therefore, a PRPE QC program should never fully rely on a prediction model but rather use it as an additional aid to increase PRPE QC efficiency and overall PRPE quality.

Additionally, the most important predictor was the variable brand. Figs. 1 and 3 support this finding. In both models, brand was essential for the models' ability to predict the QC outcome. Fig. 3 displays one of the decision trees of the RF where it is clear that the tree immediately predicts a certain garment as accepted with a probability of 87% when the brand is either brand C, D, or E. Similar to the RF model, brand was highly significant (p -value < 0.001) in the logistic regression. The criticality of the brand is apparent in the large discrepancies between the global QC outcomes of the different brands reaching up to a maximum discrepancy of 35.1% (Fig. 1). A predictor variable that was also available but not included in this study was lead equivalence, being the simulated pure lead thickness to achieve the same x-ray attenuation. The PRPE included in this study was deliberately confined to 0.50/0.25 because the EN 6331-3:2014 replaced the old EN norm from 1994 as the 1994 norm did not include fluorescence and secondary emissions. New PRPE are hence all manufactured

**Fig. 2.** Comparison of the ROC curves for both models. The AUC in both cases is 87%. The dashed line indicates no discrimination.

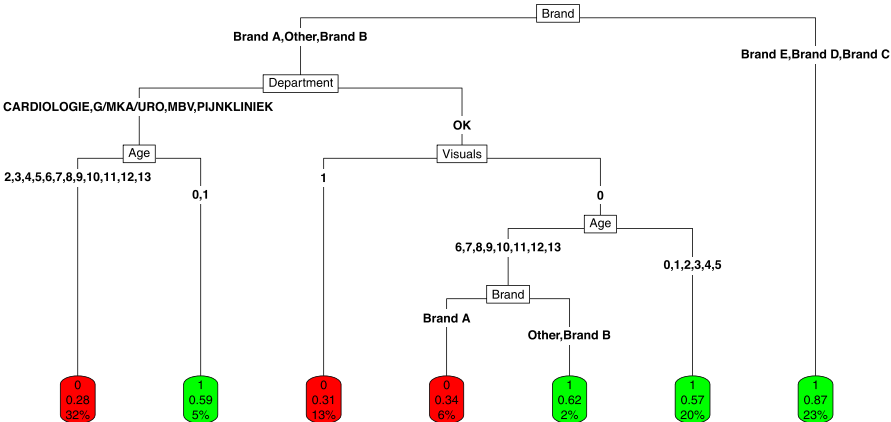


Fig. 3. Example classification tree in the RF model.

certified with a CE marking regarding the new EN norm and usually have a lead equivalence of 0.50/0.25 mm Pb. Additionally, thyroid shields were also left out from the dataset because they are a separate type of PRPE in terms of size and lead equivalence, making them less prone to defects (Kellens et al. 2022).

This study has its limitations. The dataset is based on a dataset of yearly quality controls of one hospital. Research regarding PRPE and PRPE integrity would only benefit from extra datasets from other hospitals. Unfortunately, research in

PRPE integrity is very scarce. Also, there is no regulatory standard concerning PRPE integrity assessment. Therefore, facilities develop their own policies and methodology such as rejection criteria and frequency of QC. Furthermore, it is mostly unknown what the composition of the attenuating layer of each brand is and, more importantly, how it reacts to physical stress or ionizing radiation. Future material analyses of PRPE should shed light on this. Additionally, the rejection criteria used during the data collection originate from Lambert and McKeon (2001), who established their rejection

Table 6. Overview of the false positive contradicting predictions.^a

Type	Brand	Visual defects	Department	Age	Size	Actual result	Logit prediction	RF prediction
Skirt	A	0	ST	5	M	Rejected	Pass	Rejected
Skirt	A	0	ST	4	L	Rejected	Pass	Rejected
Apron	Other	0	G/MKA/URO	-	L	Rejected	Pass	Rejected
Skirt	B	0	MI	-	L	Rejected	Pass	Rejected
Skirt	B	0	MI	8	L	Rejected	Pass	Rejected
Vest	B	1	ST	-	L	Rejected	Pass	Rejected
Vest	B	1	ST	-	L	Rejected	Pass	Rejected
Vest	B	1	ST	-	-	Rejected	Pass	Rejected
Vest	A	0	ST	8	XL	Rejected	Pass	Rejected
Skirt	A	0	ST	-	-	Rejected	Pass	Rejected
Vest	A	0	ST	8	-	Rejected	Pass	Rejected
Vest	A	0	ST	8	-	Rejected	Pass	Rejected
Vest	B	1	ST	-	-	Rejected	Pass	Rejected
Skirt	B	1	ST	-	-	Rejected	Pass	Rejected
Skirt	B	1	ST	-	-	Rejected	Rejected	Pass
Vest	A	0	ST	5	XL	Rejected	Rejected	Pass
Vest	D	1	ST	-	-	Rejected	Rejected	Pass
Vest	E	0	Cardiology	12	L	Rejected	Rejected	Pass
Skirt	B	0	ST	6	L	Rejected	Rejected	Pass
Skirt	B	0	ST	-	-	Rejected	Rejected	Pass
Vest	C	1	ST	-	-	Rejected	Rejected	Pass
Vest	A	0	ST	-	-	Rejected	Rejected	Pass
Skirt	B	0	ST	-	-	Rejected	Rejected	Pass
Apron	D	0	MI	-	-	Rejected	Rejected	Pass

^a Missing values are represented by em dashes. PC = Pain clinic, Cardiology = Catheterization lab, ST = Surgery theatre, MI = Medical Imaging, G/MKA/URO = Gastroenterology/Oral and maxillofacial surgery/Urology.

criteria based on estimated dose equivalents of medical staff and outdated radiation weighting coefficients. Furthermore, the rejection criteria are based on averted costs per person-sievert dating from 1975 (US NRC 1975a). Future rejection criteria should be updated according to the latest revision of NUREG-1530 (US NRC 2022).

CONCLUSION

To the best of our knowledge, this study is the first to predict PRPE QC outcome based on readily available predictors without the need of x-ray imaging. The results show that it is possible, up to a certain degree, to predict the quality control outcome. However, solely relying on the model would be careless and irresponsible. After all, false positives in this study are predicted passes, while the pieces would be rejected otherwise. Rejected pieces would still be worn and would expose operators to increased radiation doses. Nonetheless, the model-guided QC program could increase overall efficiency and make room for more frequent x-ray-based integrity analyses of pieces that are at risk of being rejected. This would lead to a better follow-up of the PRPE and increase overall radiation shielding quality. Future material analyses are deemed necessary to determine the composition of the attenuating layer and its association with QC outcome.

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